PAVAN KUMAR MASINA
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Senior software engineer having over 16 years of experience with expertise in end-to-end commercial software design, development and implementing modern cloud infrastructure solutions involving multi-site teams with effective interpersonal skills and team coordination capabilities.

- ✓ Broad experience in architect, design and development of enterprise, desktop and modern Azure cloud software solutions.
- ✓ Hands on experience in Azure development worked on **Azure web application, Azure storage, Azure SQL Database**, Virtual machines, **Azure AD** and notification hub
- ✓ Expertise in design and developing software solutions in **Python, C++ 14/17/20, C#.Net, Unix shell scripting, PowerShell, Core Java and Oracle PL/SQL**.
- ✓ Expertise in software design principles, algorithms and data structures, multithreading design.
- ✓ Good experience in version control using **SVN, GIT** and **CI/CD** with tools like Jenkins, Maven, Team city etc.
- ✓ Vast experience in various phases of product development with exposure to feature planning, design and implementation in various industry domains like investment banking, health care and automotive.
- ✓ Good experience in implementing credit risk models in invest banking domain.
- ✓ In depth knowledge of agile software development processes.

Education:

- **PG diploma** in embedded systems design from Center for development of advanced computing (CDAC), Bangalore, India during March 2005 to July 2005 securing B+ grade.
- **Bachelor of technology (B.Tech)** in Electronics and communications Engineering from Jawaharlal Nehru technological university, Hyderabad, India in April 2004 with First class grade.

Technical Skills:

Cloud platform	MS Azure
Programming languages	Python, C++ 14/17/20, C#.NET, PL/SQL
Databases	Oracle 19c, MS SQL server 2019/2022
Scripting languages	Bash shell scripting, PowerShell
Software frameworks	Pandas, NumPy, google test
Operating systems	UNIX, Redhat Linux, Windows server 2012 R2/2019
Software design principles	Design patterns, Object oriented design, UML, SOLID principles, Algorithms and Data structures
Protocols	SOAP, HTTP, UDDI, FTP, TCP/IP, UDP/IP, SSL
Other tools	MS office (Excel, PowerPoint, MS project), Visual studio 2017/2019/2022, PL/SQL developer, SQL server Management studio, Microsoft Visio pro

Certifications:

- Completed 1Z0-149: Oracle Database PL/SQL Developer Certified Professional certification in Feb,2024

Employment History:

Organization	**Period**	**Designation**
TATA consultancy services	Dec 2015 - Till date	Assistant Consultant
Cerner healthcare solutions Private Limited	Dec 2013 - Dec 2015	Software Architect
Robert Bosch Engineering and Business Solutions Limited	Oct 2007 - Nov 2013	Associate Project Manager

Project Summary:

Tata consultancy services

Designation: Assistant Consultant

Project: Market risk calculation system (MaRS)

Duration: Dec 2015 to till date

Client: Credit Suisse AG

Work location: Bangalore, India and Zurich, Switzerland

Responsibilities/Deliverables:

- Leading software development and maintenance activities in the market risk computing platform within the Investment Banking (IB) division, involving various complex risk calculation models such as VAR, ERC, IRC, CVA, and CCAR among others.
- Successfully implemented capital models to measure various credit risks including counterparty, international lending, Lombard, and non-Lombard using C++ and Python programming languages. This achievement aligns with the business strategic goal of standardizing various calculation models under a unified banking platform.
- Created automated unit tests using the industry-standard Google Test framework within a test-driven development (TDD) approach for critical software modules resulting in a 50% reduction in production defects.
- Contributed to the adoption of existing on-premises Windows grid compute infrastructure into the modern Azure cloud platform, leading to significant reductions in server hosting costs.
- Created detailed timeline project plans for migrating Oracle databases from version 11g to the latest 19c version with minimal downtime for business teams, as part of end-of-life (EOL) infrastructure activities.
- Implemented proof of concepts (POCs) for adopting enterprise applications into the modern Azure cloud platform as part of the department-wide cloud adoption initiative roadmap.
- Built Azure cloud infrastructure required for optimal extraction, transformation, and loading (ETL) of data from a wide variety of sources using SQL, Python, and PySpark technologies to support Quants model developer teams and data scientists during calculation model development.
- Actively involved in the migration of application security permission models to the latest AURA framework, considering important technical considerations such as caching, real-time updates, and multiple processes and applications accessing external services.

Cerner healthcare solutions private Ltd

Designation: Software architect

Project: Cerner millennium EHR product enhancements

Duration: Dec 2013 to Dec 2015

Work location: Bangalore, India

Responsibilities/Deliverables:

- Designed and developed new C#.Net UI features in the mPage framework, a component of the Cerner Millennium product.
- Ensured that all developers carry out code reviews via SonarQube as part of the code quality checks, resulting in a 10% increase in developer productivity.
- Provided guidance in troubleshooting complex problems in the Cerner Millennium electronic health record (EHR) system during live production incidents and problem calls, and recommending appropriate solutions.
- Applied refactoring to legacy Java and C++ software modules to optimize dependencies of complex modules, improve maintainability and enhance efficiency, thereby reducing future business costs associated with implementing new product features.

- Actively participated in all phases of the software development lifecycle by applying an in-depth understanding of complex healthcare industry methodologies, regulatory policies and controls.
- Managed middleware, systems and services administration, monitoring, and deployment across both test and production environments.
- Actively contributed to optimize the performance of EHR software system by analysing bottlenecks, implementing efficient algorithms and tuning system configurations.

Robert Bosch business and engineering solutions

Designation: Associate project manager

Project: ETK - Emulation test probe

Client: ETAS

Duration: Jan 2011 to Nov 2013

Work Location: Bangalore, India and Stuttgart, Germany

Responsibilities/Deliverables:

- Conducting functional analysis, timeline assessment, cost estimation, requirements allocation and interface definition studies to transform customer needs into hardware and software specifications.
- Contributed to the development of Windows application drivers and tools for Automotive ETK embedded hardware for ECU testing using C++, C#.Net, windows API technologies.
- Possess extensive experience with INCA tools utilized for Electronic Control Units (ECU) development and testing, as well as for validation and calibration of electronically controlled systems in vehicles. This experience extends to usage on the test bench or in a virtual environment on the PC.
- Developed numerous complex features for the INCA application solution using various languages, frameworks, and platforms, including C++, C#, Java, SQL, Oracle, Python, MS Visual Studio, Log4Net, BOA protocol, Wireshark, NANT, XML, XSD, XSLT, and SVN.
- Collaborated closely with clients in the planning, design, and delivery of INCA/ETK/XETK products and services.
- Created Technical specification documents with UML diagrams like Activity Diagrams and State diagrams for a better understanding of flow of module for future reference.

Project: eASEE SW configuration management tool

Client: Robert Bosch GmbH

Duration: Nov 2007 to Dec 2010

Work Location: Bangalore, India and Stuttgart, Germany

- Involved in development of customized plugins for 3rd party custom configuration management tool eASEE (product of Vector Gmbh) using COM, C++, VC++ technologies to integrate process specific requirements.
- Implemented design patterns in C++ code for better code maintainability.
- Participate in system integration testing (SIT) and user acceptance testing (UAT) to identify application errors and to ensure quality software deployment.